

Alberta Palaeontological Society

Bird-like Traits of Theropod Dinosaurs

Speaker: Darla K. Zelenitsky, PhD, Associate Professor, University of Calgary

Location: Webinar On-line Presentation

Time: April 17, 2026, 7:30 pm MDT

Abstract:

This presentation examines the evolutionary link between theropod dinosaurs and birds, emphasizing evidence from the fossil record beyond skeletal anatomy. Early research by Huxley and Ostrom highlighted skeletal similarities, while recent findings have uncovered shared characteristics in feathers, nesting behaviors, and egg biology. Central to this discussion are oviraptorid dinosaurs, a group of feathered, toothless theropods whose fossils, including eggs and nests, are well-documented in Asia. Our published work on oviraptorosaur eggs and perinates has advanced our understanding of egg formation, egg types, nest structures, reproductive strategies, and hatching behaviors. Overall, dinosaurs possess reproductive traits that are transitional between reptiles and birds, with small theropods like oviraptorids possessing the most avian-like characteristics.

Biography:

Her research focuses on dinosaur paleobiology, with particular emphasis on theropod dinosaurs and the evolutionary relationships between dinosaurs and birds. She was a primary researcher in the description of the first feathered dinosaurs discovered in the Americas, preserved in an ostrich-mimic dinosaur from Alberta, Canada. She has published numerous scientific papers on dinosaur eggs, embryos, and nests, using these fossils to explore behavioral and other biological implications. Over the past two years, she has collaborated with graduate students and colleagues worldwide to help document several major paleontological discoveries, including three new species of apex predatory theropods (Thanatotheristes from Alberta, Ulughbegsaurus from Uzbekistan, and Khankhuuluu from Mongolia), an exquisitely preserved oviraptorid embryo, a giant Cretaceous turtle embryo-egg, and the first reported soft-shelled eggs in dinosaurs.